DIPLOMA PROGRAMME PRACTICE · HL · SET B

# Biology / Paper 1A

**Question booklet · 40 marks · 120 minutes across 1A and 1B**

Answer every question. Choose one option, A–D, per question. Use a calculator. Record choices online or on separate paper.

**QUESTION 1** [1]

A DNA strand has sequence 5′-AGTCCA-3′. Which sequence is the complementary strand written in the 5′ to 3′ direction?

**A** 5′-TCAGGT-3′
**B** 5′-TGGACT-3′
**C** 5′-AGTCCA-3′
**D** 5′-UGGACU-3′

**QUESTION 2** [1]

A DNA fragment contains 300 nucleotides across both strands. Exactly 60 are adenine. Using two hydrogen bonds per A–T pair and three per G–C pair, how many hydrogen bonds join its bases?

**A** 390
**B** 420
**C** 360
**D** 780

**QUESTION 3** [1]

A short RNA strand promotes a chemical reaction and also serves as a copying template. Why is this relevant to an RNA-first hypothesis?

**A** One type of molecule could support catalysis and information storage
**B** Catalytic RNA alone establishes that early cells used the modern genetic code
**C** Any molecule that can copy information must also form its own membrane
**D** The experimental RNA establishes that early metabolism required protein enzymes first

**QUESTION 4** [1]

An enveloped virus buds from a host membrane. Which component of the new particle most directly derives from the host lipid bilayer?

**A** Its lipid envelope
**B** The sequence of a viral surface glycoprotein, specified solely by the host membrane
**C** The phage-like nucleic acid core, formed from host cholesterol
**D** The viral capsid, assembled directly from host membrane phospholipids

**QUESTION 5** [1]

A field worker identifies a specimen as Panthera leo. Which inference follows from the binomial alone?

**A** Its population is reproductively isolated from every other population of lions
**B** Its closest living relative must be Panthera pardus
**C** It belongs to the same species as every Panthera
**D** It shares its genus with Panthera pardus

**QUESTION 6** [1]

Two similar-looking plants have nucleotide differences at 6 of 400 aligned positions. A third differs from the first at 24 positions. Assuming equal rates and comparable sites, what can be inferred?

**A** The first plant is four times older than the third
**B** Exactly six mutations occurred in their entire evolutionary history
**C** The first pair has fewer observed changes per site and may share a more recent ancestor
**D** Their different habitats prove the sequence comparison invalid

**QUESTION 7** [1]

An island population of flowering plants splits into early- and late-flowering groups in the same habitat. Pollinators remain active during each period. Which change most directly promotes reproductive isolation?

**A** Both groups have identical flowering times
**B** Minimal overlap in flowering times reduces pollen exchange between groups
**C** Every plant receives more pollen irrespective of group
**D** Increased seed dispersal alone guarantees isolation

**QUESTION 8** [1]

A habitat contains 12 butterfly species before clearance and 12 after clearance. Which additional result shows a loss of biodiversity that richness alone misses?

**A** Several formerly common native species are replaced by widespread generalists
**B** Both counts are made in summer
**C** Butterflies were sampled at noon on both dates
**D** Sampling effort is identical

**QUESTION 9** [1]

A polypeptide contains 96 amino acid residues in one unbranched chain. How many water molecules were released when free amino acids formed its peptide bonds?

**A** 192
**B** 96
**C** 95
**D** 97

**QUESTION 10** [1]

A protein contains two polypeptide chains held together partly by disulfide bridges. A reducing agent breaks these bridges without hydrolysing peptide bonds. Which change is possible?

**A** The peptide backbone is converted to DNA
**B** The chains separate while their individual amino acid sequences remain unchanged
**C** Each disulfide bridge releases a codon
**D** Each chain immediately loses all amino acids containing sulfur

**QUESTION 11** [1]

A cotransporter moves glucose into intestinal cells against its concentration gradient while $Na^+$ moves down its electrochemical gradient. If the $Na^+$ gradient collapses, which prediction follows?

**A** The carrier automatically hydrolyses glucose to restore the $Na^+$ gradient
**B** Glucose must now move uphill by simple diffusion
**C** All glucose transport elsewhere in the organism stops
**D** Net uphill glucose uptake by this carrier decreases

**QUESTION 12** [1]

Lysosomal enzymes work best at a lower pH than enzymes in the cytosol. What advantage follows from a lysosomal membrane?

**A** It raises the lysosomal pH to match the cytosol during normal digestion
**B** It prevents any transfer of products from the lysosome to the cytosol
**C** It keeps substrates permanently outside while soluble enzymes digest them from within
**D** It maintains a separate chemical environment for digestion

**QUESTION 13** [1]

A chloroplast mutant retains an intact envelope and normal stromal volume but has greatly reduced thylakoid membrane area. Which capacity is most directly reduced?

**A** Housing the Calvin-cycle enzymes in their main reaction compartment
**B** Housing the majority of chloroplast DNA within the envelope
**C** Housing photosystems and associated electron carriers
**D** Maintaining the volume of stroma available for carbon fixation

**QUESTION 14** [1]

In a sealed bell-jar model of ventilation, pulling down the flexible membrane expands the balloons. What immediately drives air into them?

**A** The pressure outside the jar suddenly falls
**B** Oxygen diffusion causes the model diaphragm to contract
**C** Pressure inside the balloons falls below external pressure
**D** Carbon dioxide actively pumps air through the tube

**QUESTION 15** [1]

At identical oxygen partial pressure, fetal haemoglobin is more saturated than maternal haemoglobin. What is the significance at the placenta?

**A** Fetal haemoglobin cannot release oxygen to tissues
**B** The two blood supplies must mix directly
**C** Fetal blood can load oxygen at partial pressures where maternal haemoglobin releases it
**D** Oxygen diffuses from lower to higher partial pressure without any gradient

**QUESTION 16** [1]

A phloem source cell loads sucrose into a sieve tube. Under the pressure-flow model, which sequence follows?

**A** Water enters osmotically, increasing hydrostatic pressure and driving flow toward a sink
**B** Sucrose moves solely by individual diffusion from root to leaf
**C** Water leaves, raising pressure and driving flow toward the source
**D** Air embolisms pump sucrose through the xylem

**QUESTION 17** [1]

Calcium concentration rises in a skeletal muscle fibre. What change allows cross-bridge cycling to begin?

**A** Myosin heads convert directly into actin
**B** The Z lines disintegrate
**C** ATP is no longer involved in detachment
**D** Movement of tropomyosin exposes binding sites on actin

**QUESTION 18** [1]

Two aquatic mammals have similar shapes. Species X has twice the linear dimensions of Y. Under identical conditions, which prediction follows from geometry alone?

**A** X generates heat only at its skin
**B** X has one quarter the body volume
**C** X loses less heat per unit body volume
**D** X has twice the surface-area-to-volume ratio

**QUESTION 19** [1]

An enzyme assay compares temperatures using identical concentrations. At 55 °C the initial rate is high but activity soon falls even after transfer back to 25 °C with fresh substrate. What best explains the persistent decline?

**A** Heat altered the functional conformation of some enzyme molecules
**B** The original substrate was consumed and the fresh substrate cannot react
**C** Every enzyme molecule gained another active site
**D** Cooling permanently increases collision frequency

**QUESTION 20** [1]

An inhibitor binds an allosteric site and reduces catalytic activity. Adding abundant substrate fails to restore the original maximum rate. Which interpretation fits best?

**A** A persistent lower maximum establishes reversible competition with substrate at one active site
**B** Substrate occupancy alone cannot overcome loss of functional catalytic capacity
**C** At saturation all forms of reversible inhibition must give the uninhibited maximum
**D** The shape of the rate curve alone distinguishes covalent from noncovalent inhibition

**QUESTION 21** [1]

Equal numbers of yeast cells supplied with glucose release $CO_2$ both with and without oxygen. Which conclusion follows?

**A** The oxygen-free culture cannot regenerate $NAD^+$
**B** $CO_2$ release alone cannot distinguish aerobic respiration from alcoholic fermentation
**C** $CO_2$ production proves that oxygen was consumed
**D** Both treatments yield equal ATP per glucose

**QUESTION 22** [1]

Oxygen is removed from aerobically respiring mitochondria. What happens to electron flow and reduced NAD when oxygen is the terminal acceptor?

**A** Electron flow slows and reduced NAD accumulates
**B** Electron flow accelerates and reduced NAD disappears
**C** Oxygen removal directly hydrolyses all ATP
**D** The inner membrane becomes the terminal electron acceptor

**QUESTION 23** [1]

A chloroplast briefly receives labelled $CO_2$ in light. Which reaction initially incorporates the labelled carbon into an organic intermediate of the Calvin cycle?

**A** Splitting ATP into two triose phosphates
**B** Transfer of excited electrons from chlorophyll
**C** Photolysis of water
**D** Carboxylation of RuBP by Rubisco

**QUESTION 24** [1]

A lipid-soluble signal alters transcription in target cells. Which observation best supports direct action via an intracellular receptor?

**A** A signal–receptor complex binds a regulatory DNA sequence before mRNA increases
**B** The signal elevates a cytosolic second messenger in receptor-expressing cells
**C** A short pulse of signal rapidly opens a plasma membrane ion channel
**D** The signal raises receptor mRNA concentration without measurable binding at regulatory DNA

**QUESTION 25** [1]

At a typical chemical synapse, why does information normally pass from the presynaptic neuron to the postsynaptic cell?

**A** Both cells release identical transmitter simultaneously
**B** Neurotransmitters can diffuse in only one direction in water
**C** Action potentials cross the cleft as a continuous flow of cytoplasm
**D** Transmitter release machinery and postsynaptic receptors are arranged asymmetrically

**QUESTION 26** [1]

During the absolute refractory period, a stronger-than-usual stimulus fails to initiate another action potential at the same site. Which mechanism explains this?

**A** All ions have left the neuron
**B** Myelin becomes permeable to DNA
**C** Many voltage-gated $Na^+$ channels are inactivated and cannot yet reopen
**D** Transmitter diffuses backwards down the entire axon

**QUESTION 27** [1]

A shoot tip rests on agar in darkness. When a small agar block from beneath the tip is placed on one side of a decapitated growing shoot, that side elongates faster. A plain agar block shows no effect. What inference is best supported?

**A** The experiment determines the complete chemical structure of the substance
**B** A diffusible growth-promoting substance moved from the tip into the agar
**C** All differential growth is independent of hormones
**D** Light itself was physically stored as glucose in the agar

**QUESTION 28** [1]

Ten randomly located quadrats, each 0.25 $m^2$, contain 35 seedlings in total. At the estimated mean density, how many seedlings occupy 20 $m^2$?

**A** 700
**B** 175
**C** 70
**D** 280

**QUESTION 29** [1]

A replication system receives all four deoxyribonucleotides but lacks a template. Which property of normal DNA replication is directly lost?

**A** The number of types of deoxyribonucleotide available falls from four to two
**B** Complementary pairing that specifies the new strand sequence
**C** Incoming nucleotides lose their complementary hydrogen-bonding groups
**D** The direction of movement of DNA polymerase along its template reverses

**QUESTION 30** [1]

A replication fork exposes templates with opposite orientations. Why must one daughter strand be assembled from fragments?

**A** Helicase unwinds only one of the two strands
**B** Ligase synthesizes whole chromosomes 3′ to 5′
**C** DNA polymerase extends 5′ to 3′ while the templates run antiparallel
**D** One parental strand contains RNA bases instead of DNA bases

**QUESTION 31** [1]

Two mature mRNAs from one gene contain different combinations of exons. Which mechanism can produce the difference without a DNA mutation?

**A** Alternative DNA replication origins specify exon order in mature RNA
**B** Changing the rate of peptide bond formation determines which exons remain in an mRNA
**C** Alternative splicing of the primary transcript
**D** Different ribosomes translate the same mature mRNA using different genetic codes

**QUESTION 32** [1]

A guide RNA matches two similar loci. Cas9 cleavage is found at both. Which claim is directly weakened?

**A** The guide targets only the intended locus
**B** The cells contain double-stranded DNA
**C** RNA can contain complementary sequences
**D** Cas9 can bind a guide RNA

**QUESTION 33** [1]

At which event of meiosis is chromosome number per daughter nucleus reduced from diploid to haploid?

**A** Separation of homologous chromosomes in meiosis I
**B** DNA replication before meiosis I
**C** Pairing of bases during interphase
**D** Separation of sister chromatids in mitosis

**QUESTION 34** [1]

A regulatory DNA sequence is removed but the coding region remains intact. Protein concentration falls while protein degradation rate is unchanged. Which measurement best tests reduced transcription?

**A** The amount of mature mRNA measured at only one time point
**B** Rate of synthesis of new RNA from the gene
**C** The rate of incorporation of amino acids into the receptor polypeptide
**D** The number of receptor proteins per cell at steady state

**QUESTION 35** [1]

A plant cell in water has solute potential −0.60 MPa. Neglect gravity and assume external water potential is 0 MPa. At equilibrium, what pressure potential is needed?

**A** +1.20 MPa
**B** −0.60 MPa
**C** 0 MPa
**D** +0.60 MPa

**QUESTION 36** [1]

An investigator blocks the LH surge but maintains earlier follicular growth in a mammal. Which event is most directly prevented?

**A** Fertilization of an already released egg in a different individual
**B** Every previous rise in oestrogen
**C** Production of all oocytes during fetal development
**D** Ovulation

**QUESTION 37** [1]

A test cross gives four phenotype classes in proportions 0.41, 0.39, 0.11 and 0.09. If the two large classes are parental, what inference best fits?

**A** The recombination frequency is 80%
**B** Crossing over occurred in precisely 20% of all cells
**C** The two loci necessarily lie on different chromosomes
**D** The loci show approximately 20% recombination in these offspring

**QUESTION 38** [1]

An individual secretes too little insulin after a meal. Which response is most directly impaired?

**A** The immediate rise in glucose caused by intestinal absorption
**B** Return of elevated blood glucose toward its usual range
**C** Glycogen breakdown in the liver in response to glucagon
**D** Secretion of glucagon by pancreatic alpha cells during fasting

**QUESTION 39** [1]

Following a water load, plasma osmolarity falls. Which combination is expected in a healthy person once the response develops?

**A** More ADH and no change in water reabsorption
**B** Less ADH and more dilute urine
**C** More ADH and more dilute urine
**D** Less ADH and more concentrated urine

**QUESTION 40** [1]

In a large population meeting Hardy–Weinberg assumptions, 36% express a recessive phenotype. If 500 individuals are sampled, how many heterozygotes are expected?

**A** 300
**B** 180
**C** 240
**D** 80